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Sometimes Painful but Promising: Feasibility and Trade-offs of On-Device Language Model Inference

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The rapid rise of Language Models (LMs) has expanded the capabilities of natural language processing, powering applications from text generation to complex decision-making. While state-of-the-art LMs often boast hundreds of billions of parameters and are primarily deployed in data centers, recent trends show a growing focus on compact models—typically under 10 billion parameters—enabled by techniques such as quantization and other model compression techniques. This shift paves the way for LMs on edge devices, offering potential benefits such as enhanced privacy, reduced latency, and improved data sovereignty. However, the inherent complexity of even these smaller models, combined with the limited computing resources of edge hardware, raises critical questions about the practical trade-offs in executing LM inference outside the cloud. To address these challenges, we present a comprehensive evaluation of generative LM inference on representative CPU-based and GPU-accelerated edge devices. Our study measures key performance indicators—including memory usage, inference speed, and energy consumption—across various device configurations. Additionally, we examine throughput-energy trade-offs, cost considerations, and usability, alongside an assessment of qualitative model performance. While quantization helps mitigate memory overhead, it does not fully eliminate resource bottlenecks, especially for larger models. Our findings quantify the memory and energy constraints that must be considered for practical real-world deployments, offering concrete insights into the trade-offs between model size, inference performance, and efficiency. The exploration of LMs at the edge is still in its early stages. We hope this study provides a foundation for future research, guiding the refinement of models, the enhancement of inference efficiency, and the advancement of edge-centric AI systems.

CCS Concepts: • **Computing methodologies** → **Natural language generation**; • **General and reference** → **Performance**; **Evaluation**; • **Hardware** → **Power and energy**; • **Computer systems organization** → **Embedded systems**.

Additional Key Words and Phrases: Edge AI, Small Language Models (SLMs), Large Language Models (LLMs), Embbeded AI, Generative AI, Benchmarking

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1 Introduction

Language models (LMs) have gained significant attention in recent years, particularly following the launch of the ChatGPT service in autumn 2022 [42].

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Beyond their role in conversational AI and virtual assistants, these models are driving substantial transformations across multiple industries, including healthcare, education, entertainment, and software development [6, 46, 54, 61]. Leading-edge models like GPT-4o have established new benchmarks for AI performance, excelling in general knowledge evaluation (MMLU), mathematical reasoning (GSM8K), and code generation (HumanEval), while also supporting multimodal interactions [40]. Even fields such as IoT and edge computing are already integrating or preparing to adopt these advanced technologies. These advancements come, however, with significant computational and storage demands, which are primarily managed through cloud-based infrastructures. While cloud-hosted inference provides scalability and centralized optimization, it also introduces challenges such as increased network latency, data privacy concerns, and high operational costs.

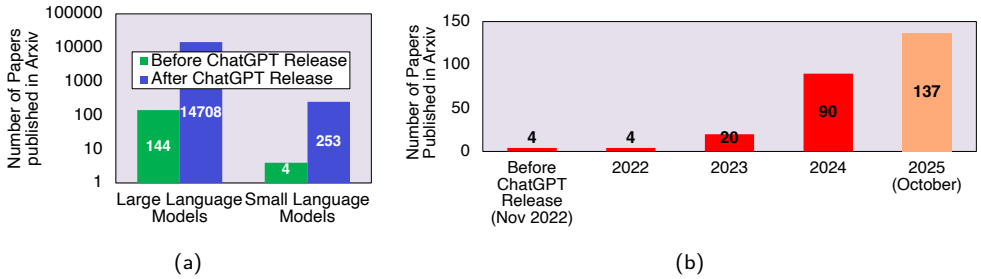

Fig. 1. Comparison of the number of research papers published on arXiv mentioning 'Large Language Model' (LLM) or 'Small Language Model' (SLM) in their title before and after the release of ChatGPT (a) and for SLM over the last years (b), illustrating the rising research interest in edge-optimized AI models.

To address these issues, there is a growing trend toward deploying LMs at the edge, paralleling earlier shifts in AI inference from centralized data centers to resource-constrained, decentralized environments. Similar to pre-LM AI workloads, efforts to enable LM execution closer to the edge are driven by the need to reduce latency, enhance reliability, and improve user experiences. This growing interest in smaller-scale models is also reflected in the research community. As shown in Figure 1, the number of papers published on arXiv mentioning "Large Language Model" in their titles skyrocketed from 184 (over a decade) to 8,480 in just a few years after ChatGPT's release. While work on Small Language Models (SLMs) remains a more niche area, its trajectory follows a similar trend, with research output rising from just 4 papers before ChatGPT's release to over 126 in early 2025. This indicates that SLMs are becoming a significant research focus, aligning with the broader push toward efficient, resource-constrained AI models. It is important to note that there is no strict definition of what differentiates an LLM from an SLM in terms of parameter count. While LLMs often exceed tens or even hundreds of billions of parameters, SLMs typically remain under 10 billion, though this boundary is not universally defined. Nevertheless, the trend toward optimizing models for edge and mobile deployment is clear, reinforcing the need to investigate the feasibility of running LMs on resource-constrained devices.

Despite this increasing interest in edge-centric LMs, however, the question remains: ***to what extent can resource-constrained devices efficiently execute LM inference?***

Recent advances in model compression [62], such as quantization and knowledge distillation, have enabled the development of more efficient LMs for edge and mobile environments. While quantization reduces memory and computational requirements by lowering parameter precision, distillation transfers knowledge to smaller models with fewer parameters, making them more suitable for resource-constrained settings. Examples include Phi-3.5 [1], Llama 3.2 [4], and a variety of emerging mobile-optimized LMs. In this respect, the rapid rise of these compact models reflects a broader shift toward efficient, resource-aware AI, as shown in Figure 2, which illustrates the increasing number of sub-4B models releases over recent years–we focus on sub-4B models to maintain clarity, as including larger models would make the visualization overly crowded.

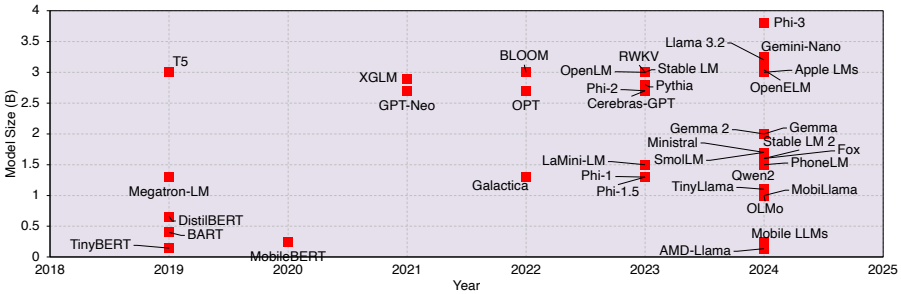

Fig. 2. Growth of sub-4B parameter language models over recent years, reflecting the increasing trend toward compact models optimized for edge and mobile deployment.

While these advancements improve feasibility for on-device inference, deploying even sub-10B models on edge devices remains computationally challenging. Unlike cloud environments, edge hardware is inherently constrained by limited memory, lower computational power, and energy efficiency concerns. Consequently, the trade-offs between model size, performance, and efficiency must be carefully evaluated to determine how well edge devices can handle LM inference in real-world applications.

To explore these trade-offs, this paper presents a comprehensive performance evaluation of generative LM inference on two distinct single-board computers (SBCs): the CPU-based Raspberry Pi 5 and the GPU-accelerated NVIDIA Jetson Orin Nano. Our study assesses key performance metrics, including memory usage, inference speed, and energy consumption, under different device configurations. Additionally, we investigate the throughput-energy trade-offs, cost implications, usability factors, and qualitative model performance. By quantifying the specific limits imposed by memory and energy constraints, this work wants to offer concrete and practical insights into the feasibility and trade-offs of running LMs at the edge.

Specifically, we explore the following research questions (RQ#):

(RQ1) What are the computational limits of running LMs on edge devices?
(RQ2) How do different model configurations (size, quantization) impact performance at the edge?
(RQ3) What are the trade-offs between CPU- and GPU-based inference on resource-constrained hardware?
(RQ4) How do system configurations (power modes, threading, etc.) affect inference efficiency, including microarchitectural-level performance metrics such as cache misses and context switches?
(RQ5) What are the usability constraints, cost implications, and qualitative trade-offs when running LM inference on edge devices?